Quantum Changepoint Detection:

A Statistical Framework for Cyber-Physical Threat Recognition

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IQIM
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C2QA
Co-design Center for Quantum Advantage

Autonomous Hamiltonian certification and changepoint detection

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1 VIRGINIA TECH 2 PHASECRAFT 3 Duke UNIVERSITY 4 Caltech

Hamiltonians

- A system that evolves over time can be described by the Schrödinger equation:

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = \boldsymbol{H} |\psi(t)\rangle$$

- $\boldsymbol{H}$ is a Hamiltonian: Hermitian linear operator
- Described by unitary evolution when H is time-independent

$$|\psi(t)\rangle = e^{-iHt/\hbar} |\psi(0)\rangle =: U(t) |\psi(0)\rangle$$

Hamiltonians: Use cases

- Scenarios requiring modeling quantum interactions:
  - Quantum chemistry
  - Material design
  - Many-body physics
  - etc.
- A promising application is to simulate these on a quantum computer
- Dynamics of quantum devices can likewise be described by Hamiltonians

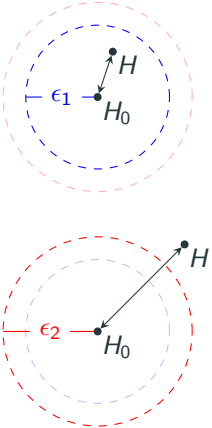
Hamiltonian Certification

Certification: Setting

- A device is promised to implement some *ideal* Hamiltonian H_0 (known)
- Instead, the implemented Hamiltonian is H
- Need to distinguish between the following cases:

$$\text{(Near):}\ \|H - H_0\|_F \leq \epsilon_1,$$
$$\text{(Far):}\ \|H - H_0\|_F \geq \epsilon_2$$

- $\|\cdot\|_F$ is the *normalized*[a] Frobenius norm
- Design an algorithm that with high probability:
  - **Accept**s in the (Near) case,
  - **Reject**s in the (Far) case
- Minimize samples and total evolution time
- Applicable to benchmarking / validation



[a] Divided by $\sqrt{d}$, which accounts for the scaling of the Frobenius norm under stochastic noise on every parameter.

Certification: Assumptions and practicality

- How does a quantum device certify itself?
- *Autonomous*: no external assistance
- *Single-qubit* state preparation and measurements
- Examples of resources this prohibits:
  - Second device / trusted ancilla,
  - Evolution $e^{-iH_0 t}$ by the ideal Hamiltonian,
  - Entangled state preparation and measurements,
  - Controlled / inverse-time evolution,
  - Trotterization
- Hamiltonians have *operator norm bounded* by M
  - n-qubit, traceless, constant bound on coefficients

Quantum state certification [GHO25]

- Algorithm for certifying quantum *states*
  1. Randomly sample $k \in \{1, \ldots, n\}$,
  2. Measure qubits 1 to $(k-1)$ in computational basis,
  3. Measure qubits $(k+1)$ to n in *decision tree* basis,
  4. Certify on qubit k
- Only uses single-qubit gates, no controlled / inverse-time evolution
- $\mathcal{O}(n\xi^{-1}\ln(1/\delta))$ samples for precision $\xi_1 = \xi/(2n)$, $\xi_2 = \xi$
- Dependence on system size n not necessary with multi-qubit gates
  - n present in sample complexity and threshold ratio
  - Future direction (out of scope)

[GHO25] Meghal Gupta, William He, and Ryan O'Donnell. *Few single-qubit measurements suffice to certify any quantum state*. 2025

Hamiltonian certification procedure

- More total parameters (compared to pure states)
- More flexibility: choose input states and measurements
  1. Sample stabilizer state $|\psi_0\rangle \in \{|0\rangle, |1\rangle, |+\rangle, |-\rangle, |i\rangle, |-i\rangle\}^{\otimes n}$,
  2. Evolve by H for time $t = cM^{-1}$,
  3. Certify using [GHO25] w.r.t. a classically-computed state,
  4. Repeat N times and perform a likelihood ratio test

- Relating distance $\|H_0 - H\|_F$ and infidelity $1 - F(\rho, \sigma)$
  - Accumulated differences described by evolution $W(t) = e^{iH_0 t} e^{-iHt}$,
  - Bound I term in Pauli decomposition,
  - Analyze 4th order Taylor expansion of $|\mathrm{tr}[W(t)]|^2$,
  - Proof uses normal ordering

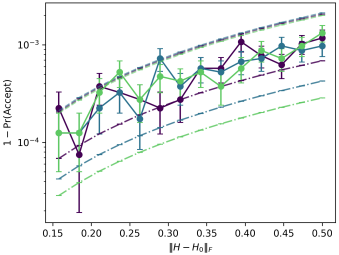
[GHO25] Meghal Gupta, William He, and Ryan O'Donnell. *Few single-qubit measurements suffice to certify any quantum state*. 2025

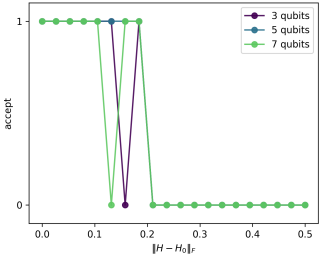
Certification: Results

- Total sample complexity and evolution time

$$N = \mathcal{O}(nM^2\epsilon^{-2}\ln(1/\delta)) \ , \quad T = \mathcal{O}(nM\epsilon^{-2}\ln(1/\delta))$$

- Tolerance thresholds $\epsilon_1 = \mathcal{O}(\epsilon/\sqrt{n})$, $\epsilon_2 = \epsilon$
- Classical computation polynomial in the number of nonzero Pauli terms $\mathbf{m}$
- Comparison to prior work [Gao+26]:
  - Linear in n; linear M factor instead of $\mathbf{m}^{3/2}$
  - n dependence removeable via [FL11] (at the cost of worse ϵ-scaling)





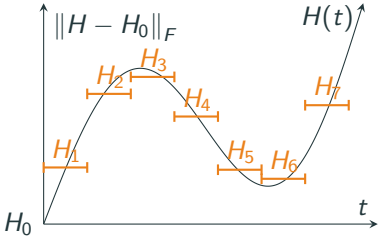
[Gao+26] Minbo Gao, Zhengfeng Ji, Qisheng Wang, Wenjun Yu, and Qi Zhao. "Quantum Hamiltonian certification". *Proceedings of the 2026 Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*. 2026

[FL11] Steven T. Flammia and Yi-Kai Liu. "Direct fidelity estimation from few Pauli measurements". *Physical Review Letters*. 2011

Hamiltonian Changepoint

Time-dependent Hamiltonians / Quantization

- Hamiltonian $H(t)$ can in general vary with time
- Analysis becomes more complicated
- Quantize: sequence of Hamiltonians $H_1, H_2, \ldots$
- **Frame challenge:** does this lose information?
  - High-frequency components missed
  - Bound spectrum of the Hamiltonian

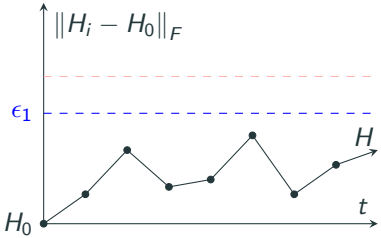


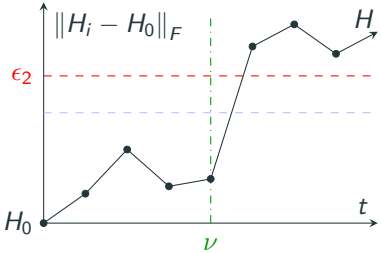
Changepoint: Setting

- A change could happen at any sequence index ν
- Need to distinguish between the following cases:

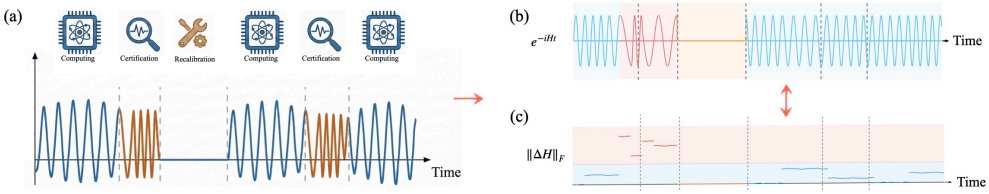
(No change): $\|H_i - H_0\|_F \leq \epsilon_1$ for all $i \in \{1, 2, \ldots\}$,
(Change): $\|H_i - H_0\|_F \geq \epsilon_2$ for all $i \in \{\nu + 1, \nu + 2, \ldots\}$

- Design an algorithm that with high probability:
  - **Keeps running** in the (No change) case,
  - **Terminates** *rapidly* in the (Change) case
- Minimize detection delay $\overline{\mathbb{E}}$;
- Maximize runtime to false alarm $\mathbb{E}_\infty$





Changepoint: Applications



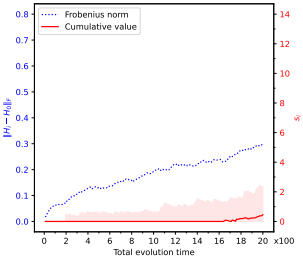
- Noise / anomaly / threat detection in an *online* fashion
- Recalibration problem:
  - Changes in $H(t)$ represent accumulated miscalibrations,
  - An (expensive) recalibration procedure can be run to reset $H(t)$ to H_0,
  - When to trigger recalibration?

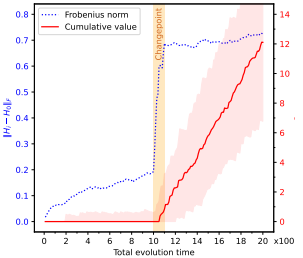
Adversarial framework

- No assumptions on the stochasticity of the miscalibrations
  - Loss to an environment,
  - Collected by an eavesdropper,
  - Manipulated by an adversary
- Adversary could have full information about the used architecture / protocol
- Models various types of attacks
  - Extraction of sensitive information,
  - Manipulation of outcomes,
  - Denial of service

CUSUM [Pag54]

- Classical procedure for continuous (online) inspection
  - Monitor for changes / anomalies in time series data
  - Terminate rapidly following a sudden change
- Initialize a cumulative score $S_0 = 0$. At step i:
  1. Receive time-series data X_i,
  2. Compute log-likelihood ratio $Z_i = \ln \frac{dQ}{dP}(X_i)$,
  3. Update $S_i = \max\{0, S_{i-1} + Z_i\}$,
  4. Terminate if $S_i \geq h$
- Score stays near zero pre-change;
- Increases linearly post-change





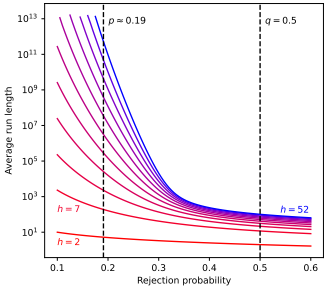
[Pag54] Ewan S. Page. "Continuous inspection schemes". *Biometrika*. 1954

CUSUM: Optimal scaling [Lor71]

- Equivalent to a parallel series of open-ended ratio tests:
  - Terminate when change is e^h times more likely
  - Predicts MLE for past change location
- Known optimal scaling for delay

$$\overline{\mathbb{E}}[\max\{0, N-\nu\}] \geq \frac{\ln \mathbb{E}_\infty[N]}{D(Q \mid P)} \cdot (1 + o(1))$$

- Robust over classes of distributions
  - Analysis via least-favorable distributions [MF19]



[MF19] Timothy L. Molloy and Jason J. Ford. "Minimax robust quickest change detection in systems and signals with unknown transients". *IEEE Transactions on Automatic Control*. 2019

[Lor71] Gary Lorden. "Procedures for reacting to a change in distribution". *The annals of mathematical statistics*. 1971

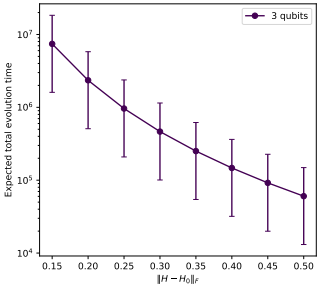
Changepoint: Results

- Use $P = \mathrm{Ber}(\xi/2n)$, $Q = \mathrm{Ber}(\xi/n)$
- Relation between detection delay and false alarm runtime

$$\overline{\mathbb{E}}[\max\{0, N-\nu\}] \leq \mathcal{O}\left(\frac{nM^2 \ln \mathbb{E}_\infty[N]}{\epsilon^2}\right),$$

$$\overline{\mathbb{E}}[\max\{0, T-t\nu\}] \leq \mathcal{O}\left(\frac{nM \ln(M\mathbb{E}_\infty[T])}{\epsilon^2}\right).$$

- Tolerance thresholds $\epsilon_1 = \mathcal{O}(\epsilon/\sqrt{n})$, $\epsilon_2 = \epsilon$



Conclusion / Future directions

- *Autonomous* miscalibration / anomaly / threat detection
- Certification for offline settings
- Online changepoint detection
- Quantum algorithm sensitivity linear in operator norm M
- Classical simulation complexity linear in sparsity $\mathbf{m}$

- Dependence on the number of qubits n
- Analysis of spectrum bounds for time-dependent Hamiltonians
- Integration of changepoint methods with quantum data
- Adaptive choice of input states

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Thank you!

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